Akoizo Original Practice - 2027 Rotation Form 2 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Parallel resistors share…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage

B. Always current

C. Always power

D. Always resistance

Correct: A

2. (2 points) A voltmeter connects…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. As an intentional source short

B. Only instead of every wire

C. Only to ground alone

D. Across two nodes

Correct: D

3. (2 points) NOT(A AND B) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. A OR B

B. (NOT A) AND B

C. A AND B

D. (NOT A) OR (NOT B)

Correct: D

4. (2 points) An open switch can have…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Infinite real power

B. Always zero voltage

C. Consumed charge

D. Voltage with zero current

Correct: D

5. (2 points) Divider inference needs…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Zero source voltage

B. No known component

C. The same series current

D. Equal resistor values

Correct: C

6. (4 points) Give half-adder equations.

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Sum = A XOR B; carry = A AND B. (4 points).

7. (4 points) Find 100 Ω parallel 100 Ω.

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: 50 Ω. (4 points).

8. (4 points) What does meter loading mean?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: The instrument's finite impedance changes the quantity being measured. (4 points).

9. (4 points) Why is logical one not always 5 V?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Physical signal ranges depend on device technology and supply. (4 points).

10. (4 points) Give a power check.

Unit: Diagrams, nodes, series, parallel, and Ohm’s law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: Source VI should equal the sum of resistor I^2R losses in the ideal network. (4 points).

11. (2 points) A 8 V ideal battery drives a 4 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 4 and 6 ohms. Find equivalent resistance.

answer: 10 ohms (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , ohms.

13. (2 points) Two identical 8 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 4 ohms (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 8 V. Find power.

answer: 16 W (2 points).

Numerical target 16; absolute tolerance 0.032. Accepted units: , W.

15. (2 points) A node has incoming currents 5 A and 2 A and one outgoing current. Find the outgoing current.

answer: 7 A (2 points).

Numerical target 7; absolute tolerance 0.014. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

answer: In the ideal model the reverse-biased diode blocks current; a forward-biased conducting diode permits conventional current from anode toward cathode. Real leakage and forward voltage require a stated device model. (2 points).

19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).